



Structure & Syllabus for Semester-V
Integrated M.Sc. (Computer Science) Programme
offered in
Department of Computer Science
Gujarat University
2025 – 2026

**As per NEP 2020 CURRICULUM AND CREDIT FRAMEWORK FOR
UNDERGRADUATE PROGRAMMES, UGC**

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of

Education Department, Govt. of Gujarat

Semester - V

(Integrated M.Sc. (Computer Science) Programme)

SKILL ENHANCEMENT COURSES

Course Name: Advanced Python Programming – Practicals

Course Code: SEC-356

Credits: 2

Course Outcomes:

The aim of this course is to enable students to

- **CO1:** Use various libraries for implementing advance concepts such as high order functions, Network programming, etc. in Python
- **CO2:** Gain basic insight of programming help them to deal with data and prepare data to be fit for application of Machine Learning and Deep Learning algorithms

Prerequisites:

Fundamentals of Python

Contents:

Unit No.	Particulars	Credits	Time
1.	Unit Title: Understanding of High Order Functions, Network Programming and Data Handling Techniques <i>High Order functions:</i> map, reduce and filter, Function Decorators, Generators, <i>Network Programming:</i> Introduction of Network and server – client concept, Knowing IP addresses, URL parsing, Webpage downloading from Internet, Socket programming, Multithreading <i>Data Handling techniques:</i> Data sets, Data handling using NumPy, Pandas, Matplotlib and Seaborn libraries, Creating our Own Modules in Python	01	30 Hrs.
2.	Unit Title: Exploratory Data Analysis (EDA), Scikit-learn, Model Evaluation Techniques <i>EDA:</i> Handling Missing Values, Scaling data, Encoding Categorical Variables, Feature Selection, Feature Extraction, Outlier Removal, Balancing data and performing Feature Engineering, <i>Scikit-learn (sklearn) library:</i> Linear and Logistic Regression, K-Nearest Neighbours (KNN), Support Vector Machines (SVM), Decision Trees, and Random Forests, <i>Model Evaluation using metrics:</i> Accuracy, Precision, Recall, F1-score, R2score, MAE, MSE, RMSE, Hyperparameter Tuning with GridSearchCV and RandomizedSearchCV	01	30 Hrs.

References:

1. Rao N.R., “Core Python Programming”, Dreamtech Publication India
2. Chun W., “Core Python Programming”, Prentice Hall India
3. Lee K.D., “Python Programming Fundamentals”, 2nd Edition, Springer International
4. Halterman R., “Fundamentals of Python Programming”, Southern Adventist University
5. Gutttag J.V., “Introduction to Computation and Programming Using Python”, Prentice Hall India

Accomplishments of the student after completing the course:

At the end of the course, students will be able to

- Implement Functions and advanced programming concepts using Python
 - Perform Network programming using Python programming language
 - Apply data handling techniques using Python to collect data, manipulate, pre-process data, analyse data and visualize data
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